ITM SLS BU

School of Computer Science, Engineering & Technology, Vadodara

B.Tech Semester-5 AY 2026-2027,Winter-2026

Advanced Web Development

Regular Final Viva-Voce Submission

As part of the **Advanced Web Development (AWD)** subject, students are required to complete and demonstrate the following tasks based on the project developed under the **Project-Based Learning (PBL) component using a JavaScript framework**.

The tasks must be relevant to the your project and will be evaluated during the regular final viva-voce examination.

Task-1. Comparative Technology Study

Compare below **technologies, frameworks, tools, or approaches** relevant to your project.

Examples:

- REST API vs. GraphQL
- MongoDB vs. MySQL
- React vs. Angular
- JWT-based authentication vs. session-based authentication

Deliverable: A comparative report covering the following aspects:

- Introduction to both technologies
- Key features and differences
- Advantages and limitations
- Suitable use cases
- Justified conclusion explaining which technology is more appropriate for your project and why

Task-2. Security Audit Report

Conduct a basic security audit of your PBL project to identify potential vulnerabilities and suggest appropriate preventive measures.

The audit should consider the following security aspects, wherever applicable:

- SQL/NoSQL injection
- Cross-Site Scripting (XSS)
- Weak authentication and authorization
- Insecure password storage
- Inadequate input validation
- Other relevant security risks identified in your project

Deliverable: A security audit report containing:

- Identified vulnerabilities and their descriptions
- Risk classification (High, Medium, or Low)
- Potential impact of each vulnerability
- Recommended preventive measures
- Security improvements implemented or proposed in the project

Task-3. Implementation and Demonstration of Security Libraries

Justify the use of the following Node.js libraries in your PBL project or a suitable sample application:

1. **Helmet.js:** To configure HTTP security headers and improve application security.
2. **bcrypt:** To securely hash passwords before storing them in the database and to verify passwords during login.
3. **Joi:** To validate user input and ensure that incoming data meets the required validation rules before processing.

Deliverable: A suitable demonstration with relevant code snippets or syntax, explanations, and execution results showing how these libraries are integrated and used.

Submission Instructions

- All three tasks must be based on or relevant to your PBL project.
- **Students should prepare the necessary PPT and word document for the viva-voce examination.**
- Students must be able to explain the concepts, implementation, advantages, limitations, and security considerations of their chosen technologies and libraries.
- The investigate work should demonstrate a clear understanding of the concepts rather than merely copying code from external sources.
- **The viva-voce examination will be conducted in the same groups formed for the submission of the PBL project developed using a JavaScript framework.**

Note: Students are expected to demonstrate both their theoretical understanding and practical implementation during the final viva-voce examination.

Best Wishes

Happy Coding!!!!

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